

Hardware article

TinZr: A compact wireless ESP32-C3 platform for multi-modal physiological sensor integration and data acquisition

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ABSTRACT

Microcontrollers have become central to wearable technologies. As these systems continue to evolve, both form factor and functional integration have become critical design considerations. In this study, we present *TinZr*, a compact, wireless, and open-source ESP32-C3-based microcontroller that serves as a central hub for interfacing with external sensing modules, data logging, and wireless transmission. The *TinZr* integrates an inertial measurement unit (IMU), an onboard battery management circuit, Wi-Fi and Bluetooth Low Energy, a microSD card reader, and multiple GPIOs supporting digital, analog, I²C, SPI, and UART interfaces. The complete system measures only 25 mm × 25 mm, making it well-suited for wearable applications. All production files, including printed circuit board (PCB) design, firmware, and a 3D model of a wearable enclosure, are publicly available. Three primary use cases for *TinZr* are demonstrated: (a) multi-device communication and coordination, (b) wireless IMU data acquisition, and (c) simultaneous multi-modal physiological signal acquisition using external electrocardiography (ECG) and photoplethysmography (PPG) sensing modules. The *TinZr* supports both real-time data streaming via Wi-Fi/BLE and local logging to a microSD card. Local logging makes the *TinZr* suitable for extended research experiments and field studies where continuous wireless connectivity may not be available. An accompanying Arduino library, *TinZrConnect*, is provided to facilitate programming and integration with existing systems. Overall, the combination of compact design, wireless flexibility, and open-source accessibility makes *TinZr* a versatile solution for various sensing applications.

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Specifications table

Hardware name	TinZr
Subject area	• Electrical and biomedical engineering • Neuroscience • Swarm robotics • Educational tools and open source alternatives to existing infrastructure
Hardware type	• <i>Measuring physical properties and in-lab sensors</i> • <i>Field measurements and sensors</i> • <i>Electrical and biomedical engineering</i> • <i>Multi-functional microcontroller platform</i> • <i>Compact and wearable embedded controller</i> • <i>Wireless data acquisition and communication system (Wi-Fi/BLE)</i> • <i>Real-time signal processing</i> • <i>Multi-modal physiological signal acquisition and sensor interfacing (ECG, PPG, IMU)</i> • <i>Digital communication protocols support (I²C, SPI, UART, GPIO)</i>
Closest commercial analog	• Seeed XIAO ESP32-C3 • ESP32-C3 Mini • DFRobot Beetle ESP32-C6 • Adafruit GEMMA v2 • Teensy 4.1
Open source license	CERN Open Hardware Licence (OHL) v2.0 Permissive (hardware), GNU General Public License (GPL) v3.0 (software)
Cost of hardware	\$65/unit (including shipping and customs duties)
Source file repository	https://doi.org/10.5281/zenodo.21892119
OSHA certification	US002796
UID	

1. Hardware in context

Wearable sensing platforms are gaining popularity for their ability to deliver non-invasive, continuous health insights to athletes, patients, and everyday users — while evolving to incorporate sophisticated on-device sensors and analytics. The global wearable sensors market was valued in 2024 at USD 1.4 billion and is projected to grow to USD 6.3 billion by 2033, at a compound annual growth rate (CAGR) of 18.3% over 2025–2033 [1,2]. This surge is largely driven by growing demand for telehealth, long-term monitoring, real-time health tracking, and military applications.

Physiological signal monitoring has become increasingly widespread, with wearable systems now employed by athletes, patients, soldiers, and everyday users to enable continuous health and performance tracking [3–6]. Consequently, many everyday wearables (e.g., watches) now go beyond basic functionality by including onboard sensing hardware, such as accelerometers for activity and motion tracking [7]; photoplethysmography (PPG) sensors for heart rate [8] and oxygen saturation calculation [9]; electrocardiography (ECG) sensors for cardiac rhythm monitoring [10]; temperature sensors for body or skin thermal profiling [11]; and even biochemical sensors for measuring variables like glucose [12], for richer health and activity monitoring [4,13].

Wearable systems for physiological monitoring increasingly depend on microcontroller units (MCU) and platforms that provide tight integration, compact form factors, and support for multi-modal sensing. The continued miniaturization and increasing computational capability of modern MCUs have made it possible to embed real-time sensing, processing, and wireless communication within small, battery-powered systems. Additionally, such MCUs offer sufficient local processing power, robust data logging, and versatile interfaces for auxiliary modules such as inertial, optical, and electrophysiological sensors.

Several open hardware and research platforms have emerged to address these challenges. For instance, *CogWatch* [14] is an open-source multi-sensor wrist platform that integrates heart rate, respiration, skin conductance, and temperature sensing, to study physiological and behavioral responses in daily activities. *CogWatch* uses the *Adafruit QT Py ESP32-S2* MCU [15] for data acquisition and measures around 29 mm × 38 mm (just the PCB without any sensor enclosures). *H-Watch* [16] extends these capabilities into a wristwatch-like platform that incorporates multiple radios and energy harvesting, maintaining ultra-low power consumption for long-term wearable deployment. The main purpose of *H-Watch* is early COVID-19 detection. *BioGAP-Ultra* [17] is a modular edge-AI platform that supports synchronized acquisition of EEG, EMG, ECG, and PPG signals with on-device processing and flexible form factors. The *BioGAP-Ultra* firmware runs on the nRF5340 SoC using the Zephyr RTOS, and the accompanying Android software suite enables background data reception, real-time visualization, some parameter configuration, and data storage. Finally, the *Multi-Sensor Wearable Health Device (MWHd)* framework [18] uses the *CC2640R2F Cortex®-M3* [19] MCU for real-time heart rate and peripheral oxygen saturation (SpO₂) monitoring from PPG signals, temperature, and motion, coupled with wireless data transmission

Table 1Features and price comparison between some commercially available controllers and *TinZr* in terms of dimensions, GPIOs, interfaces, and price.

	<i>Seeed XIAO ESP32-C3</i> [20]	<i>DORHEA ESP32-C3 Mini</i> [21]	<i>DFRobot Beetle ESP32-C6</i> [22]	<i>Adafruit GEMMA v2</i> [23]	<i>PJRC Teensy 4.1</i> [24]	<i>TinZr</i>
Dimensions (mm)	21 x 17.8	22 x 18	25 x 20	28 (diameter)	61 x 18	25 x 25
Nbr. GPIOs	11	11	13	3	55	10
Nbr. Analog Pins	4	4	7	1	18	5
Nbr. Digital Pins	11	11	13	3	55	10
I ² C	Yes	Yes	Yes	Yes	Yes	Yes
SPI	Yes	Yes	Yes	Yes	Yes	Yes
UART	Yes	Yes	Yes	No	Yes	Yes
On-board MicroSD Card Reader	No	No	No	No	Yes	Yes
On-board Battery	Yes	No	Yes	No	No	Yes
Management						
On-board QWIIC	No	No	No	No	No	Yes
Connector						
On-board IMU	No	No	No	No	No	Yes
On-board Controllable RGB LED	No	No	No	No	Yes	Yes
On-board Push Button	Yes (Reset & Boot)	Yes (Reset & Boot)	Yes (Reset & Boot)	Yes (Reset)	Yes	Yes (Reset & Boot/Control)
Unit Price (in USD) (January, 2026)	~ \$5 – 10	~ \$8 – 12	~ \$7 – 10	~ \$9.95	~ \$30 – 50	~ \$65

and mobile application feedback. Together, these platforms illustrate the state-of-the-art in wearable microcontroller integration, where the convergence of low-power embedded processing, miniaturization, and open hardware design has enabled compact, reconfigurable, and affordable solutions for physiological signal monitoring.

Some of the above-mentioned platforms demand significant expertise to program or extend at the driver and firmware level. However, there are compact and flexible MCUs that can be easily programmed via the Arduino ecosystem — making them highly accessible for rapid prototyping, research deployments, and wearable device development. For example, we list several widely used controllers such as *Seeed XIAO ESP32-C3* [20], *DORHEA ESP32-C3 Mini* [21], *DFRobot Beetle ESP32-C6* [22], *Adafruit GEMMA v2* [23], and *PJRC Teensy 4.1* [24]. In general, most of these controllers are compact and thus suitable for wearable applications. The *Seeed XIAO ESP32-C3* is very compact and offers Wi-Fi/BLE connectivity with onboard battery management, though it lacks a microSD card reader and a dedicated on-board QWIIC connector for plug-and-play I²C peripheral expansion. The *DORHEA ESP32-C3 Mini* does not include battery management, an on-board QWIIC connector, or an SD card interface. The *DFRobot Beetle ESP32-C6* similarly lacks an SD card reader and an on-board QWIIC connector. The *Adafruit GEMMA v2* is extremely small and lightweight but does not include a charging unit, SD card reader, or an on-board QWIIC connector. The *PJRC Teensy 4.1* includes an SD card reader, making it well-suited for data logging, but it lacks Wi-Fi/BLE and is physically bulkier than the other options.

In this study, we introduce the *TinZr*, which builds on existing designs by providing a fully wireless controller in a compact 25 mm × 25 mm form factor, combining sensing, data logging, and communication in an open-source package. It uses the ESP32-C3 MCU and integrates an inertial measurement unit (IMU) comprising a 3-axis accelerometer and 3-axis gyroscope, battery management circuitry, Wi-Fi and BLE 5.0 connectivity, a microSD card reader, and GPIOs supporting digital, analog, I²C (it provides an on-board QWIIC connector), SPI, and UART. While *TinZr* can directly acquire analog signals through its onboard ADC, many physiological sensing applications require dedicated sensor modules or analog front-end circuitry to provide signal conditioning, amplification, filtering, and digitization. In such configurations, *TinZr* serves as the central acquisition, coordination, data logging, and communication hub for the connected sensing devices. A detailed specification comparison between the commercially available controllers and the *TinZr* is available in Table 1. Importantly, the *TinZr* can be programmed via the Arduino ecosystem, lowering the barrier to use. This functional density enables multimodal physiological signal acquisition, on-device processing, and flexible data handling (wireless streaming or local logging). All production files for the *TinZr*, including PCB designs and 3D enclosure models for wearables applications, are publicly available at <https://doi.org/10.5281/zenodo.21892119>. The onboard microSD reader supports long-duration, high-resolution logging, essential for field or extended experiments where continuous wireless transmission may

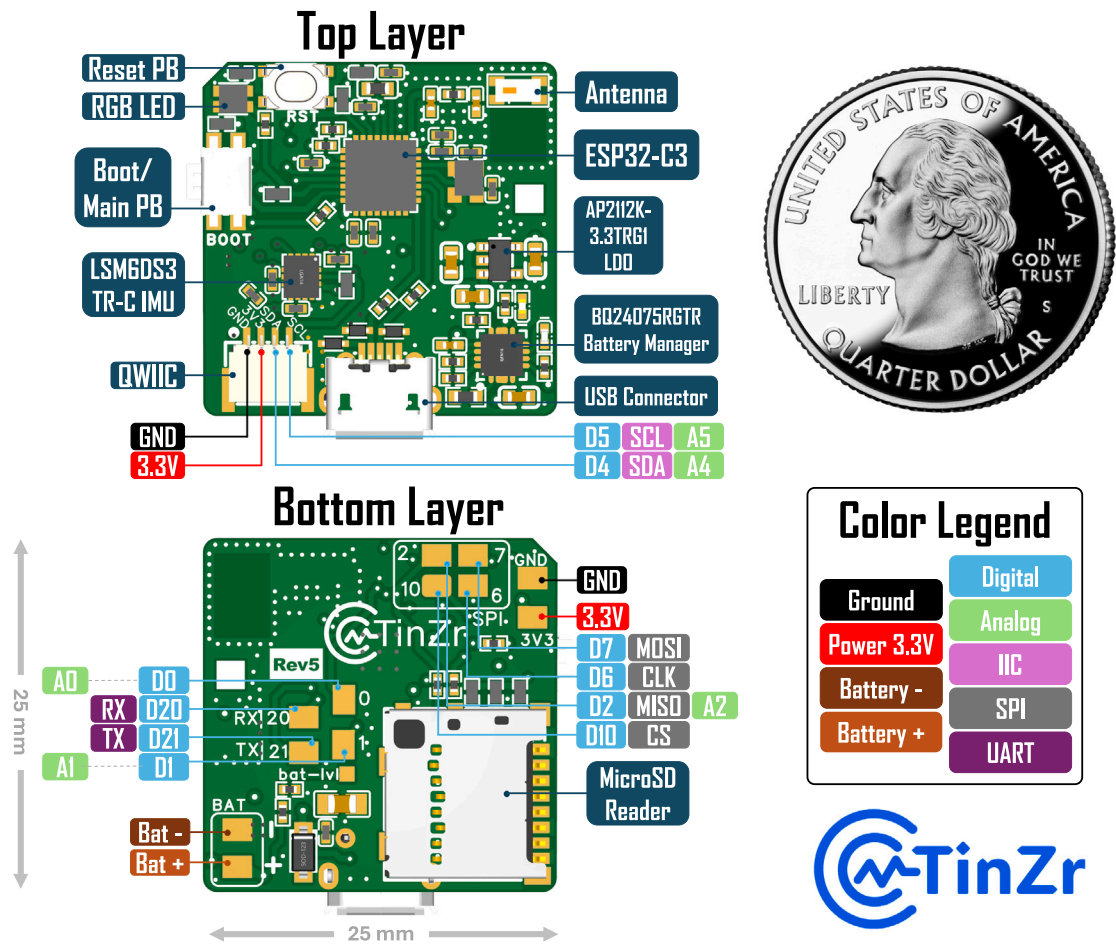


Fig. 1. *TinZr* pinout diagram showing the available GPIOs, communication interfaces, power rails, and key functional connectors.

not be feasible. Wireless connectivity via Wi-Fi and BLE remains available when real-time streaming is needed. We provide an accompanying Arduino library, *TinZrConnect*, to facilitate programming and integration of *TinZr* with existing systems.

Overall, *TinZr* offers a low-cost and compact alternative that surpasses lighter platforms by combining local storage, wireless flexibility, and standardized sensor interfacing in a single module.

2. Hardware description

Our repository, available at <https://doi.org/10.5281/zenodo.21892119>, provides the following materials:

- raw and editable schematics and PCB layout files used in the design of the *TinZr* ESP32-C3 Rev5; PCB layouts are provided in .eprj format for EasyEDA Pro [25], as well as in .kicad_pcb/.kicad_sch format for KiCad [26];
- a 3D model of the wearable enclosure, designed to house the *TinZr* device and battery, provided in .step format and compatible with FreeCAD [27] and other CAD software;
- exported manufacturing files, including Gerber files, pick-and-place data, and bills of materials (BOM) for the PCBs, as well as .stl files for 3D printing the wearable enclosure.

Fig. 1 illustrates the *TinZr* pinout along with the top and bottom layers of the PCBA. Fig. 2 shows the 3D model of the wearable *TinZr* enclosure, illustrating the exploded and assembled views used to house the PCBA and battery. The major functional blocks and embedded components of the system are described in the following subsections.

2.1. ESP32-C3 development board

The central component of the *TinZr* is an ESP32-C3-based microcontroller module, labeled “ESP32-C3” in Fig. 1. The ESP32-C3 module integrates a single-core RISC-V processor with native 2.4 GHz Wi-Fi and Bluetooth Low Energy (BLE) connectivity,

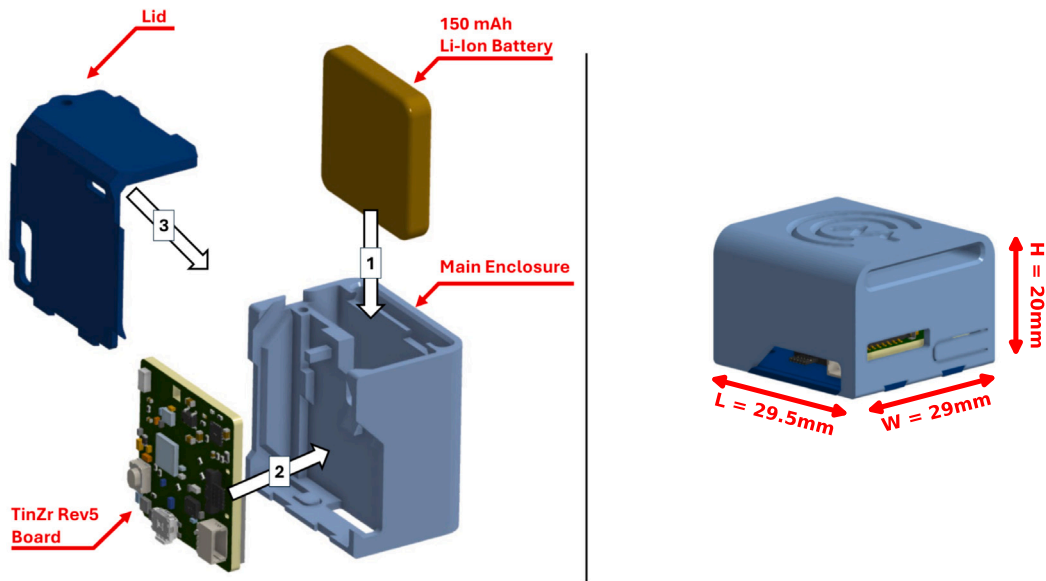


Fig. 2. 3D model of the *TinZr* wearable enclosure illustrating the mechanical design used to enclose the PCBA and battery for wearable applications. The exploded view (left panel) shows the individual enclosure components, including the main enclosure, top lid, battery lid, *TinZr* Rev 5 PCBA, and 150 mAh Li-ion battery, while the assembled view (right panel) shows the fully assembled enclosure (length = 29.5 mm, width = 29 mm, and height = 20 mm). Dashed orange arrows in the exploded view indicate the assembly directions and relative placement of each component during enclosure assembly.

providing a compact and power-efficient solution well suited for wearable and portable applications. The processor operates at clock frequencies of up to 160 MHz, offering sufficient computational performance for real-time control, coordination, and wireless communication tasks.

The ESP32-C3 module provides a sufficient number of general-purpose input/output (GPIO) pins to support the required digital and analog interfaces, including I²C, SPI, and UART communication buses, while maintaining low power consumption during both active operation and deep sleep modes. Although the ESP32-C3 module offers fewer GPIOs than larger development boards such as the Teensy 4.1, the pin allocation in *TinZr* was carefully designed to prioritize the interfaces and peripherals most relevant to physiological signals coordination, wearable sensing, and wireless data transfer. As a result, all core system functionalities are supported without reliance on external multiplexing or expansion hardware.

The ESP32-C3 module is positioned centrally on the *TinZr* PCBA, with its USB interface accessible at the board edge to facilitate firmware development, debugging, and power delivery (Fig. 1).

Overall, the ESP32-C3 module provides a compact, wireless, and power-efficient computational core that enables *TinZr*'s functionality while supporting scalable, open-hardware development.

2.2. Inertial measurement unit (IMU)

Motion sensing in the *TinZr* platform is provided by an STMicroelectronics LSM6DS3TR-C inertial measurement unit (IMU), labeled “LSM6DS3TR-C IMU” in Fig. 1. The LSM6DS3TR-C unit integrates a 3-axis accelerometer and a 3-axis gyroscope in a compact, low-power package, enabling continuous monitoring of linear acceleration and angular velocity.

The accelerometer supports selectable full-scale ranges up to ± 16 g, while the gyroscope supports angular rate measurements up to ± 2000 dps, allowing the device to capture both fine-grained motion and high-dynamic activities. The IMU supports output data rates up to several kilohertz and communicates with the ESP32-C3 module via the I²C bus, providing reliable and low-latency motion data acquisition.

The LSM6DS3TR-C unit was selected for its low power consumption, small footprint, and proven reliability in wearable and embedded applications. In the context of *TinZr*, IMU data can be used to support motion-aware sensing, artifact detection, and coordination tasks, particularly in mobile and wearable experimental settings.

2.3. Battery management and power regulation

Power management in the *TinZr* platform is handled by a Texas Instruments BQ24075RGTR battery management integrated circuit, labeled “BQ24075RGTR Battery Manager” in Fig. 1. The BQ24075RGTR is a single-cell lithium-ion/lithium-polymer (Li-ion/Li-Po) charger integrated circuit (IC) with power-path management, enabling simultaneous battery charging and system operation.

The device supports USB-compatible input sources and provides regulated power to the system while safely managing battery charging, including preconditioning, constant-current, and constant-voltage charging phases. Integrated safety features, such as thermal regulation, overvoltage protection, and charge termination, ensure reliable and safe operation in wearable and portable applications.

Power-path management allows *TinZr* to operate directly from an external power source when available, while automatically switching to battery power when disconnected, without interrupting system operation. This capability is particularly important for long-duration experiments and mobile use cases, where uninterrupted operation and safe battery handling are critical. The BQ24075RGTR was selected for its compact footprint, low external component count, and proven reliability in battery-powered embedded systems.

The charging behavior of the BQ24075RGTR unit in *TinZr* is configured using three external resistors connected to the “ISET”, “ILIM”, and “TMR” pins. The fast-charge current is set by a resistor on the “ISET” pin. With an 8.2 k Ω resistor, the resulting charge current is approximately 100 mA, which is well matched to the 150 mAh single-cell lithium-polymer (Li-Po) battery [28] used with the *TinZr* and corresponds to a charge rate of approximately 0.67C.

The maximum input current drawn from the external power source is limited via the “ILIM” pin to ensure compatibility with USB power supplies while supporting simultaneous system operation and battery charging. With a 3 k Ω resistor connected to “ILIM”, the input current is limited to approximately 520 mA, providing sufficient headroom for system operation while remaining within typical USB power constraints.

In addition, a resistor connected to the “TMR” pin sets the safety timeout for the charging process. With a 46.4 k Ω resistor, the precharge safety timer is approximately 37 min, and the fast-charge safety timeout is approximately 6.2 h, providing protection against abnormal charging conditions. These resistor values were selected specifically for the 150 mAh battery [28] used with the *TinZr*, and can be adjusted (by desoldering the onboard resistors and replacing them with desired ones) to accommodate larger-capacity batteries by increasing the allowable charge and input currents, while maintaining manufacturer-recommended charge rates and thermal limits.

The battery is soldered directly to the rear side of the PCB at the pads labeled (+) and (−), ensuring correct polarity and compact mechanical assembly.

2.4. On-board controllable RGB LED

TinZr includes an addressable RGB status LED (“NeoPixel”-class), labeled “RGB LED” in Fig. 1. This type of LED integrates red, green, and blue emitters together with a small control driver in a single package, allowing the LED color and brightness to be digitally controlled from a single microcontroller GPIO pin.

The RGB LED can be programmed to provide a visual indication of device state during development and deployment (e.g., BLE advertising/connection state, recording status, or error conditions). Since the LED can draw non-negligible current at high brightness, *TinZr* firmware limits brightness and disables the LED during low-power operation when appropriate. This feature preserves battery life while retaining a simple, user-facing diagnostic interface.

The RGB LED is a fully software-controllable indicator intended for general system feedback and debugging. *TinZr* also includes dedicated power and charging LEDs (bottom right corner of the top layer in Fig. 1) that operate independently of firmware, providing a visual indication of device power and battery charging status.

2.5. MicroSD card interface

Local data storage in the *TinZr* platform is provided by an on-board microSD card reader, labeled “microSD Reader” in Fig. 1. The microSD interface allows sensor data, timestamps, and system logs to be stored directly on removable flash media, enabling fully standalone operation without reliance on a continuous wireless connection.

The microSD card communicates with the ESP32-C3 module via the SPI bus, providing a simple and widely supported interface for embedded systems. This design choice ensures broad compatibility with standard microSD cards and commonly used embedded file systems, while maintaining predictable timing and resource usage. The SPI-based implementation also simplifies integration alongside other peripherals on the shared communication bus.

The inclusion of local storage is particularly important for long-duration recordings, field deployments, and wearable experiments where wireless bandwidth or connectivity may be limited or unavailable. By offloading data logging to the microSD card, *TinZr* supports reliable data capture without requiring continuous wireless transmission. We validated data logging using a 32 GB microSDHC card (Silicon Power Elite, FAT32-formatted).

2.6. QWIIC connector

The *TinZr* platform includes an on-board QWIIC-compatible expansion connector, labeled “QWIIC” in Fig. 1, providing a standardized interface for external peripheral integration. The QWIIC connector exposes the I²C bus along with power and ground, enabling rapid connection of compatible sensors and modules without soldering.

By leveraging the I²C interface, the QWIIC connector allows additional devices — such as environmental sensors, auxiliary motion sensors, or other low-power peripherals — to be easily integrated into the system. This design supports modular expansion while minimizing wiring complexity and reducing the risk of incorrect connections during prototyping or deployment.

The inclusion of a QWIIC connector enhances the flexibility of the *TinZr* platform by enabling rapid hardware extensions without modifying the core PCB design. This feature is particularly useful for experimental and research-oriented applications, where sensor configurations may evolve over time or vary between deployments.

2.7. Reset and user push buttons

TinZr includes two on-board push buttons: a dedicated reset button, labeled “Reset PB”, and a multifunction push button, labeled “Boot/Main PB” in Fig. 1. The reset button provides a hardware reset of the ESP32-C3 module, allowing the system to be restarted without cycling power during development or troubleshooting.

The main button serves a dual purpose. When used in conjunction with the reset button, it can be used to place the ESP32-C3 module into bootloader mode for firmware flashing (see ‘Program’ in Section 6.1). During normal operation, the main button can be repurposed by the firmware as a general user input, enabling application-specific functionality such as mode selection, event marking, or user interaction.

The main button is implemented as right-angle (90°) tactile switch. This orientation allows the button to be actuated from the side of the enclosure rather than the top surface of the PCB, simplifying mechanical integration and improving usability in compact enclosures. The right-angle design is particularly advantageous for wearable applications, where low profile, accessibility, and protection against accidental presses are important considerations.

2.8. Micro-USB connector

The *TinZr* platform includes a micro-USB connector, labeled “USB Connector” in Fig. 1. It provides both power input and a wired communication interface to the ESP32-C3 module. This connector is used for battery charging via the on-board power management circuitry as well as for firmware flashing, debugging, and development during device configuration.

When connected to an external power source, the micro-USB interface supplies power to the system and enables charging of the on-board Li-Po battery through the BQ24075 battery management IC. During development, the same interface allows firmware to be uploaded to the ESP32-C3 module and supports serial communication for debugging and status monitoring.

The micro-USB connector was selected for its compact form factor, mechanical robustness, and wide availability of cables, making it well suited for both early-stage development and enclosed wearable applications. Its placement at the edge of the PCB facilitates straightforward access when the device is integrated into a protective enclosure.

Given *TinZr*’s flexibility, reprogrammability, and support for a wide range of external sensors and communication interfaces, it can be adapted to many research applications both within and beyond wearable physiological monitoring. We envision its principal use cases as follows, although this list is by no means exhaustive:

- **Wearable and portable sensing:** *TinZr* can serve as a compact, battery-powered platform for acquiring motion, physiological, environmental, or other sensor data in laboratory, ambulatory, and field-based experiments.
- **Custom multi-sensor data acquisition:** External sensing modules can be integrated through I²C, SPI, UART, analog, or digital interfaces, allowing researchers to develop application-specific and multi-modal sensing systems without redesigning the central controller.
- **Standalone and wireless data collection:** Sensor data can be wirelessly streamed in real time over BLE or Wi-Fi, or stored locally on the onboard microSD card, enabling both interactive laboratory experiments and longer-duration recordings where continuous wireless connectivity may not be available.
- **Multi-device communication and coordination:** Multiple *TinZr* units can communicate with a central host and be controlled individually or collectively, enabling coordinated multi-device sensing, simultaneous measurements across multiple devices, and swarm-style experimental applications.

Design filename	File type	License	Location of the file
<i>TinZr</i> Rev 5 PCB	raw schematics, *.svg & *.kicad_sch; PCB sketches, *.kicad_pcb; EasyEDA Pro source, *.eprj; manufacturing files, *.grb & *.csv	CERN OHL-P 2.0	https://doi.org/10.5281/zenodo.21892119
Enclosure 3D designs	source, *.step; manufacturing, *.stl	CERN OHL-P 2.0	https://doi.org/10.5281/zenodo.21892119
BOM of all components	*.csv	CERN OHL-P 2.0	https://doi.org/10.5281/zenodo.21892119
Firmware code	Arduino IDE source, *.ino; Python-based GUIs, *.py	GNU General Public License, version 3 (GPLv3)	https://github.com/ludvikalkhoury/TinZr.git

3. Design files summary

- *TinZr* Rev5 PCBs: design files of *TinZr* Rev5 PCB are available. These are the raw and editable EasyEDA Pro files (.eproj format) as well as KiCAD PCB files (.kicad_pcb format). Also available are manufacturing files that can be used directly to produce the PCBs. These are Gerber (.grb), pick and place (.csv), and bill of materials (BOM) (.csv) files.
- Enclosure 3D designs: raw editable 3D design files (in .step format) are available for the *TinZr*'s wearable enclosure. Also are available the corresponding .stl files that can be used directly in slicing software to generate a g-code for a 3D printer.
- BOM of all components: a BOM is available that contains all the components we used to build the *TinZr* Rev5 PCB.
- Firmware code: an Arduino library, *TinZrConnect*, is available, with examples that can be directly updated and run on the *TinZr*.

4. Bill of materials (BOM)

We provide a detailed and comprehensive BOM in the *TinZr*'s Zenodo repository, listing all electronic components along with their part numbers, manufacturers, and supplier IDs.

The cost of making the *TinZr* Rev5 is around \$65 per unit. This per-unit cost assumes a minimum quantity of 5, reflecting the minimum quantity required by the PCB manufacturing company, and include shipping and current customs duties (January 2026).

5. Build instructions

Step-by-step instructions are provided, on how to manufacture the *TinZr*.

5.1. PCB manufacturing

As previously mentioned, all the files needed in the PCB manufacturing process are publicly available. These files are Gerber, Pick-and-Place, and BOM, and they can be found in the Zenodo repository. We have used JLCPCB (<https://jlcpcb.com/>)¹ to manufacture the boards since it provides a wide variety of base materials and colors and other PCB specifications. We designed and built the *TinZr* Rev 5 as a 4-layer PCB, using a standard quality PCB with flame retardant (FR-4) as the base material. We picked a PCB thickness of 1.6 mm with a hot air solder leveling (HASL) (with lead) surface finish. We also chose the copper weight on the top and bottom to be 1 oz/ft². We used two ground planes (top and bottom).

5.2. Wearable enclosure 3D printing

We provide a 3D model (.stl files) for an enclosure designed to fit the *TinZr* Rev5's PCBA as well as a 150 mAh Li-Po battery that power the *TinZr*.

The wearable enclosure was printed using a local Creality Ender-3 3D printer (Fused Deposition Modeling (FDM)), using PLA filaments. The quality of the print was acceptable for our purposes. In case the user requires a higher resolution, we suggest using a higher quality alternatives such as Multi-Jet Fusion (MJF) printing, which could result in a higher-quality product (but comes at a higher price).

5.3. Component soldering

For our design, we used the mounting and soldering services provided by JLCPCB to assemble and solder all components on the *TinZr* board. The only exception was the main push button, which was not available through JLCPCB at the time of ordering and was therefore soldered manually.

5.4. Assembly

The *TinZr* Rev 5 is fully functional out-of-the-box and does not require any assembly for basic operation. However, when integrating the system into a wearable enclosure, the following steps should be followed to properly assemble the wearable unit.

- (1) Solder the battery positive (+) and negative (−) leads to the corresponding battery terminals located on the bottom layer of the *TinZr* board (Fig. 3, Panel A).
- (2) Insert the battery into the designated battery compartment and place the *TinZr* board onto its designated mounting platform within the enclosure (Fig. 3, Panel B).
- (3) Clip the snap-fit lid onto the main enclosure to secure the assembly (Fig. 3, Panel C). The wearable unit is now fully assembled and ready for use.

¹ Other PCB manufacturers offer comparable services — for example, DKRed (<https://www.digikey.com/en/resources/dkred>), OSH Park (<https://oshpark.com/>), and PCB Way (<https://www.pcbway.com/>), to name a few.

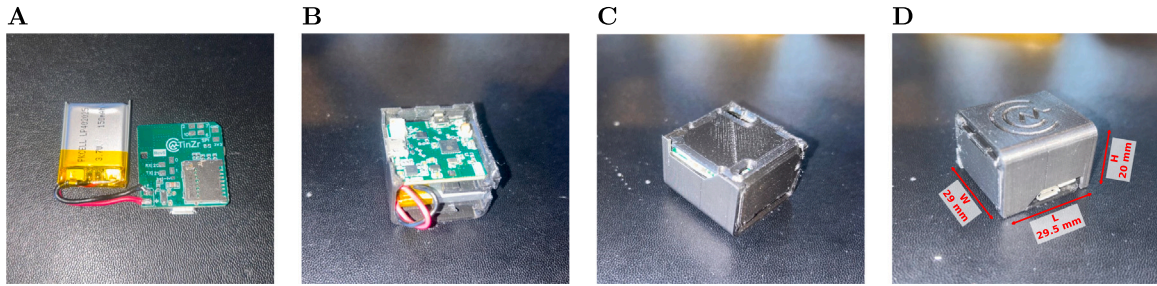


Fig. 3. Step-by-step illustration on how to assemble the *TinZr* Rev 5 board with the wearable 3D enclosure (panels A–C), following the steps described in Section 5.4. Panel D shows the fully assembled device (length = 29.5 mm, width = 29 mm, and height = 20 mm).

Panel D of Fig. 3, shows the completed *TinZr* Rev 5 wearable unit after all assembly steps have been performed.

TinZr provides a battery-level monitoring feature. Battery voltage is measured via the *bat-lvl* pad, which must be soldered to the Pin 1 pad (see the bottom layer of Fig. 1); this pin is designated in the *TinZr* firmware as the default battery-sensing input. However, it can be repurposed for other functions when battery-level sensing is not required. This connection is required for the battery level to be reported correctly by the example firmware included in the repository.

5.5. Programming

The *TinZr* Arduino library, *TinZrConnect*, can be downloaded directly from the *TinZr* GitHub repository <https://github.com/ludvikalkhoury/TinZr.git>. In addition to the Arduino library, *TinZr* provides a custom Arduino Board Manager package with full support for the *TinZr* Rev 5 hardware, enabling seamless installation and configuration within the Arduino IDE. The *TinZrConnect* library includes a collection of example sketches designed to test and demonstrate the full functionality of the *TinZr* Rev 5 board.

For instance, the *TinZrCore* sketch provides a basic demonstration of core hardware features, including the main push button, status LED, and battery level monitoring (assuming the battery-level pad has been soldered to Pin 1 pad as described in Section 5.4).

The *TinZrComCenter* sketch establishes a centralized communication framework in which multiple *TinZr* devices connect to a single host system acting as a communication hub. In this architecture, each *TinZr* operates as an independent node that can transmit data and receive control messages over Wi-Fi. The central hub manages all connected devices and can selectively broadcast commands to all *TinZr* units or address individual devices independently. This design enables coordinated multi-device operation and is well suited for distributed or swarm-style applications, such as coordinated multi-device sensing or collaborative robotics.

Another example, *TinZrWearableSD*, enables communication between one or more *TinZr* devices and a central PC over Bluetooth Low Energy (BLE), allowing the PC to control recording start and stop events while enabling local logging of inertial measurement unit (IMU) data (accelerometer and gyroscope) as well as red and infrared photoplethysmographic (PPG) data when a MAX30102 sensor [29,30] is connected via QWIIC.

Both *TinZrComCenter* and *TinZrWearableSD* are accompanied by graphical user interfaces (GUIs), along with the underlying Python source code, allowing users to operate the system without requiring extensive experience in firmware or software development. In later sections of this paper, we demonstrate the capabilities of the *TinZr* Rev 5 using *TinZrComCenter* and *TinZrWearable* as representative example use cases.

6. Operation instructions

6.1. User instructions

The *TinZr* Rev 5 operates similarly to other microcontrollers in the ESP32 family. Here we follow a simple three-step workflow: **Power**, **Program**, and **Deploy**.

- (1) **Power:** The *TinZr* Rev 5 can be powered either by connecting it to a host computer via USB or by using a connected rechargeable battery for standalone or wearable operation. When powered through USB, a white status LED illuminates, indicating that the board is receiving power. When a battery is connected and charging, a green charging-status LED remains illuminated and turns off once charging is complete. When operating solely on battery power, no LEDs are illuminated. The power and charging status LEDs are located to the left of the USB connector at the same vertical level (see Fig. 1, top layer).
- (2) **Program:** The *TinZr* Rev 5 can be programmed with any ESP32-compatible firmware, including user-developed sketches or example programs provided with the *TinZrConnect* library. Programming is performed using the Arduino IDE after selecting the appropriate board and serial port. In some cases, the board must be placed into bootloader mode manually. This is accomplished by pressing and holding the main button, briefly pressing and releasing the reset button while continuing to hold the main button, and then releasing the main button. Once the firmware upload is complete and the program begins execution, the device is ready for use.

- (3) **Deploy:** After programming, the *TinZr* Rev 5 can be deployed either while connected via USB or as a fully standalone device powered by the onboard battery. In deployed operation, the device can stream sensor data, communicate wirelessly using BLE or Wi-Fi, or operate autonomously depending on the loaded firmware. This flexibility allows the *TinZr* to function both as a development platform and as an embedded wearable system.

6.2. Safety considerations

The *TinZr* Rev 5 is a low-voltage, battery- and USB-powered microcontroller platform intended for embedded sensing, wearable prototyping, and wireless and local data acquisition. The device does not make direct electrical contact with the wearer, nor does it generate or apply electrical stimulation or excitation signals. As such, when used as intended, the *TinZr* presents minimal inherent electrical risk.

When deployed in wearable or human-adjacent applications, the *TinZr* is typically powered solely by its onboard battery and communicates wirelessly via BLE or Wi-Fi. This battery-only, wireless mode of operation significantly reduces the risk of electrical shock associated with mains-powered systems.

If the *TinZr* is connected to external sensors or peripherals that may contact a human subject, it is the user's responsibility to ensure that those components meet appropriate electrical safety and isolation standards. The platform does not provide medical-grade electrical isolation and should not be considered a standalone biomedical acquisition device. When used with human-connected sensors, the safety of the overall system depends on the design and validation of the complete measurement chain, including the external analog front-end circuitry, sensors, power supplies, grounding strategy, and any required electrical isolation. Users are responsible for ensuring that the complete system complies with applicable safety standards and institutional requirements for human-subject research. The *TinZr* Rev 5 has not been tested, certified, or approved for clinical or medical use, and should therefore be used exclusively for research, prototyping, or non-clinical applications unless additional safety certification is obtained.

As with most open-source hardware platforms, the *TinZr* is provided without warranty, and its use is undertaken at the user's own risk. The authors disclaim all liability to the extent permitted by applicable law.

7. Validation and characterization

To validate the operation and characterize the performance of the *TinZr* Rev 5 platform, we conducted several complementary validation and characterization tests. These tests were designed to evaluate the system across key intended use scenarios, including (a) multi-device communication and coordination, (b) real-time sensor (IMU) data acquisition, (c) simultaneous multi-modal physiological signal acquisition, (d) power consumption and battery life, and (e) wireless communication performance.

7.1. Multi-device communication and coordination

Multi-device communication and coordination were validated using the *TinZrComCenter* framework. In this configuration, multiple *TinZr* devices simultaneously connect to a central host system (a personal computer) that functions as a communication hub. The firmware supports bi-directional communication over TCP/IP, enabling both *TinZr*-to-hub and hub-to-*TinZr* messaging, as well as broadcast commands addressed to all connected *TinZr* units. This test case demonstrates the *TinZr* platform's ability to support coordinated, multi-node operation, enabling use cases such as distributed sensing, coordinated data acquisition, and swarm-style control architectures. For demonstration purposes, we deployed the Arduino-based firmware located at:

```
TinZr/libraries/TinZrConnect/examples/TinZrComCenter/TinZrComCenter.ino
```

First, the user should connect the PC to the same Wi-Fi network to which the *TinZrs* are intended to connect. In the firmware, the user should provide the Wi-Fi SSID and password, the PC's IP address, and a hostname for the *TinZr* onto which the code is flashed. Here, we upload the firmware to two *TinZr* devices, referred to as *TinZrBlack* (black-colored enclosure) and *TinZrGold* (gold-colored enclosure). Next, we launch the companion Python-based graphical user interface (GUI)² located at:

```
TinZr/libraries/TinZrConnect/examples/TinZrComCenter/TinZrComCenter.py
```

After both devices successfully connected to the hub, a broadcast command was issued to set the onboard RGB LEDs to blue by sending the command **LED 0 0 255 20**, which specifies red, green, and blue color intensities of 0, 0, and 255, respectively, with a low brightness level of 20. As shown in panel A of Fig. 4, the RGB LEDs on both *TinZr* devices simultaneously turned blue, confirming correct broadcast communication and successful broadcast command execution. Additionally, we can increase the brightness level to 150 and LEDs on both *TinZr* boards get brighter (panel B of Fig. 4).

² We also provide an executable file that installs the GUI on the PC without the need to run any Python codes. This file is available in the repository under the *TinZrComCenter* directory and as part of the repository release.

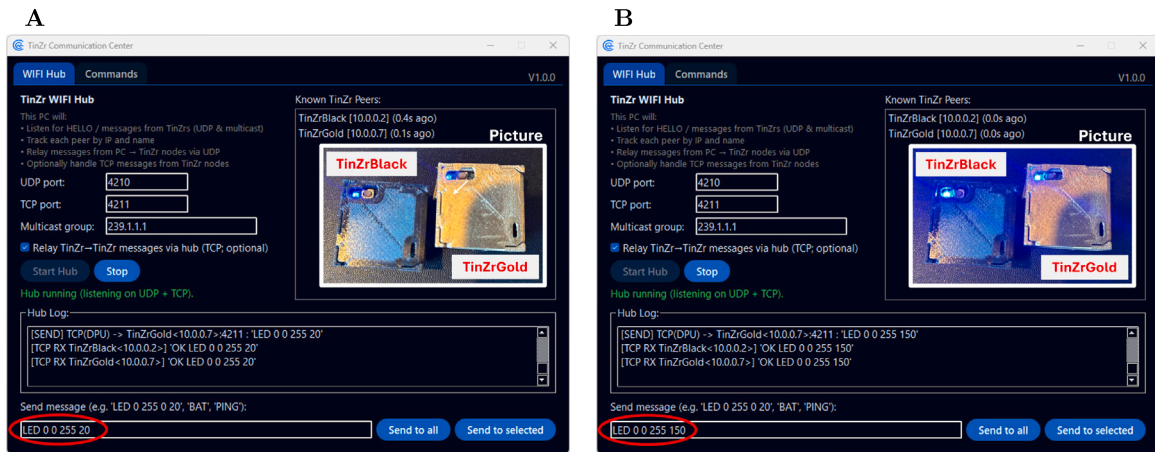


Fig. 4. Screenshot of the *TinZr* Communication Center Python-based GUI used for multi-device control over Wi-Fi, alongside a photograph of two *TinZr* devices (*TinZrBlack* and *TinZrGold*). Panel A demonstrates LED control at a low brightness setting (brightness level 20), while panel B shows the same devices driven at a high brightness setting (brightness level 150).

7.2. Real-time sensor data acquisition

We also demonstrated end-to-end wirelessly-controlled acquisition of inertial data using the *TinZrWearableSD* configuration. In this mode, the *TinZr* device streams inertial measurement unit (IMU) data acquired from the LSM6DS3TR-C sensor, including triaxial accelerometer and gyroscope signals.³ Control of recording operations, including start and stop commands, is handled through an accompanying Python-based graphical user interface (GUI) that communicates with the *TinZr* devices over BLE, while all sensor data are logged locally to the onboard microSD card. For demonstration purposes, we deployed the wearable firmware located at:

```
TinZr/libraries/TinZrConnect/examples/TinZrWearableSD/TinZrWearableSD.ino
```

The user should specify a hostname for the *TinZr* before flashing the firmware. For the microSD validation presented in this work, the firmware was configured with a sampling interval of 4 ms, corresponding to a target sampling rate of 250 Hz. Data were written to a 32 GB FAT32-formatted microSDHC card as comma-separated values (CSV) files in the */TinZrLogs* directory. Each CSV file included a metadata header containing the device name, subject identifier, PC start timestamp, sampling interval, target sampling rate, sensor detection status, and column units. The recorded variables included elapsed time, red and infrared PPG measurements from the MAX30102 (when available), and three-axis accelerometer and gyroscope measurements from the onboard IMU. Specifically, the logged columns were *t_ms*, *red_nA*, *ir_nA*, *ax_g*, *ay_g*, *az_g*, *gx_dps*, *gy_dps*, and *gz_dps*. The firmware also recorded whether the PPG sensor was detected; if unavailable, the red and infrared PPG values were logged as zero. After the firmware is flashed onto the board, we launch the companion Python-based GUI⁴ located at:

```
TinZr/libraries/TinZrConnect/examples/TinZrWearableSD/TinZrWearableSD.py
```

For illustration purposes, we collected IMU data under three conditions and report representative snapshots of the streamed sensor data in Fig. 5.

In the first condition, data were collected while the participant sat still while holding the *TinZr* in the right hand (panel A of Fig. 5). Next, the data were then collected during repetitive hand up-and-down motion at a frequency of 50 bpm (beats per minute) (panel B of Fig. 5). Finally, the motion was repeated at a frequency of 100 bpm (panel C of Fig. 5). The hand-motion frequency was controlled using a metronome.

In the absence of motion, the IMU signals show minimal activity. When motion is introduced, the IMU signals capture the corresponding periodic movement (panel B of Fig. 5). This effect becomes more frequent, with a higher rate of oscillations, as the motion frequency increases from 50 bpm to 100 bpm (panel C of Fig. 5).

³ This firmware also supports the addition of a photoplethysmographic (PPG) sensor, MAX30102 [29,30], when the optical module is present. The MAX30102 interfaces with the *TinZr* board via an I²C connection using the onboard QWIIC connector. When no MAX30102 sensor is connected, the corresponding PPG channels are reported as zero-valued signals. An example of red and infrared PPG signals collected from the MAX30102 sensor is available in Section 7.3

⁴ Similar to the previous example, we also provide an executable file that installs the GUI on the PC without the need to run any Python codes. This file is available in the repository under the *TinZrWearableSD* directory, and as part of the repository release.

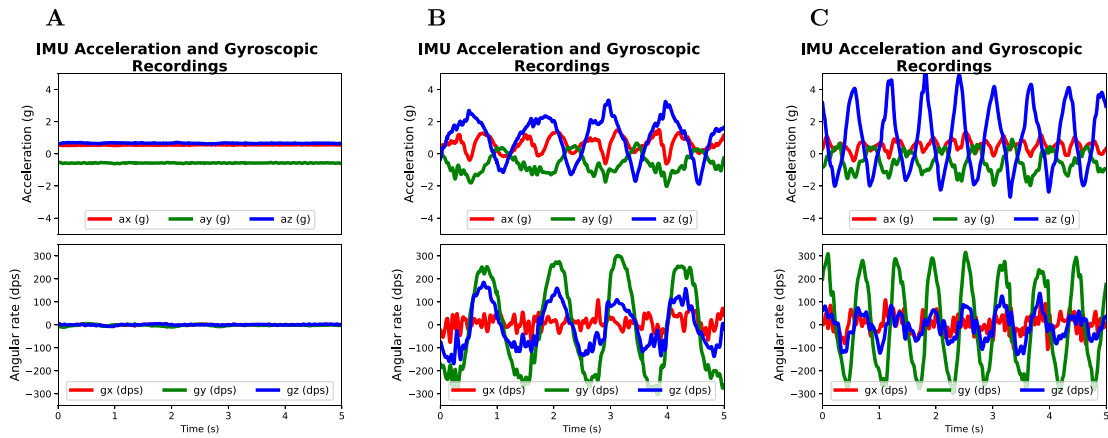


Fig. 5. IMU recordings during three scenarios; panel A, under no motion; panel B, during repetitive hand up-and-down motion at a frequency of 50 bpm; panel C, during repetitive hand up-and-down motion at a frequency of 100 bpm.

7.3. Multi-modal physiological data acquisition

To demonstrate *TinZr*'s capability to serve as a central hub interfacing with external physiological sensing modules for multi-modal physiological sensing, electrocardiography (ECG) and photoplethysmography (PPG) signals were measured simultaneously. The ECG signal was collected from a custom single-lead ECG front-end based on the MAX30001 [31], while red and infrared PPG signals were collected from the MAX30102 [29,30] optical sensor. Both external sensing modules interfaced directly with the *TinZr* platform for data acquisition and wireless transmission. The MAX30102 and MAX30001 were selected as representative examples, and *TinZr* can also interface with other compatible external physiological sensing modules.

During the recording, *TinZr* acquired the ECG and dual-wavelength PPG signals concurrently and transmitted the data wirelessly (via BLE) for visualization and storage. Because the sensors operate at different native sampling rates, the data streams were packetized using a common transmission framework in which each packet included timing and sample-counter information. This allowed the received signals to be aligned during post-processing and enabled simultaneous visualization of ECG, red and infrared PPG waveforms.

Fig. 6 shows a representative 10 s recording of simultaneously acquired physiological signals. The blue ECG trace exhibits periodic cardiac cycles with distinguishable QRS complexes, while the red and infrared PPG signals shown as red and black traces, show waveforms corresponding to peripheral blood volume changes. This example showcases the *TinZr*'s ability as a multi-modal physiological data acquisition platform.

7.4. Power consumption and battery life

Power consumption was characterized under representative *TinZr* operating modes to estimate battery life during standby, wireless communication, and physiological signal acquisition. Current was measured manually using an ammeter connected in series between the *TinZr* board and a 150 mAh single-cell Li-Po battery. Estimated battery runtime was calculated by dividing the battery capacity (150 mAh) by the measured average current consumption (mA). We characterized the power consumption and estimated runtime for the following modes:

- (1) **Sleep Idle:** In the sleep Idle condition, *TinZr* remained powered with Wi-Fi, BLE communication, sensor acquisition, and data streaming disabled, while the ESP32-C3 was placed in deep sleep without periodic wakeups. This condition represents the minimum standby current of the device when sensing and wireless communication are inactive.
- (2) **Active Idle:** In the active Idle condition, *TinZr* remained powered with the ESP32-C3 CPU active, while Wi-Fi, BLE communication, sensor acquisition, and data streaming were disabled. This condition represents the baseline current draw of the powered device before enabling wireless communication or physiological signals sensing.
- (3) **BLE Advertising:** In the BLE advertising condition, *TinZr* enabled the BLE radio and periodically transmitted advertising packets to allow wireless discovery. Sensor acquisition and BLE data streaming were disabled, and no central device connection was used. This condition quantifies the additional current required for wireless discoverability relative to active Idle.
- (4) **BLE Streaming:** In the BLE streaming condition without sensors, *TinZr* maintained a BLE connection and transmitted packets at fixed output rates of 50, 100, and 200 Hz. Sensor acquisition was disabled. This condition isolates the current contribution of BLE connected data transmission.

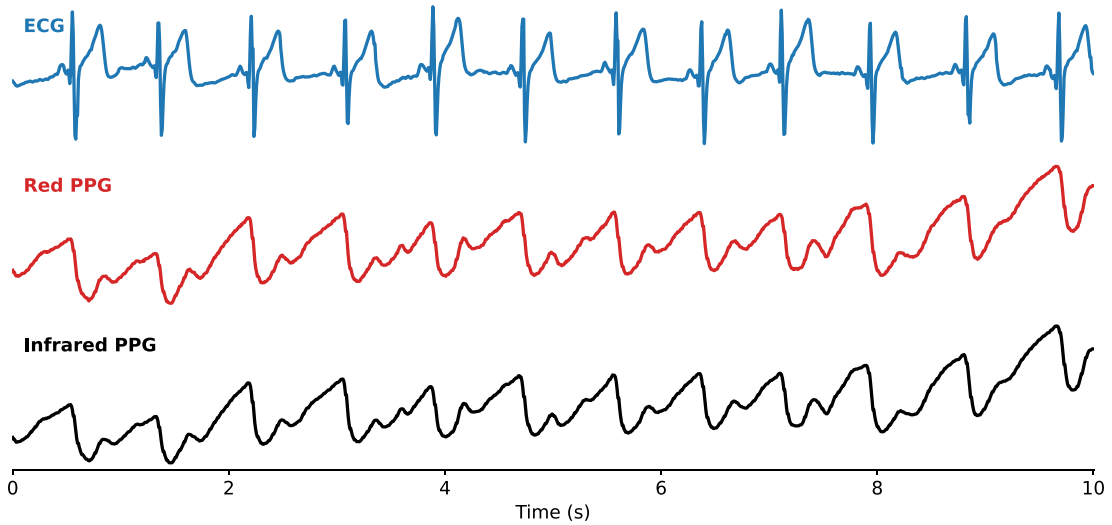


Fig. 6. Multi-modal physiological signals collected simultaneously. The ECG waveform, shown in blue, was acquired from a custom ECG front-end based on the MAX30001. Dual-wavelength PPG signals were acquired using a MAX30102 optical sensor, with the red PPG waveform shown in red and the infrared PPG waveform shown in black.

- (5) BLE + MAX30102: In the PPG acquisition condition, *TinZr* streamed red and infrared PPG signals from the MAX30102 over BLE. Measurements were performed at 100 and 200 Hz to quantify the combined current contribution of optical sensing and BLE transmission.
- (6) BLE + MAX30001: In the ECG acquisition condition, *TinZr* streamed a single-lead ECG waveform from the MAX30001 over BLE. Measurements were performed at 128 and 256 Hz to quantify the current required for ECG sensing and BLE streaming.
- (7) BLE + MAX30102 + MAX30001: In the combined sensing condition, *TinZr* acquired PPG and ECG signals simultaneously and transmitted synchronized BLE packets. Because the sensors support different sampling rates, the BLE packet transmission rate was matched to the lower sensor rate for each configuration. Specifically, BLE packets were transmitted at 100 Hz when the MAX30102 and MAX30001 were sampled at 100 Hz and 128 Hz, respectively, and at 200 Hz when they were sampled at 200 Hz and 256 Hz, respectively.
- (8) Wi-Fi Transmission: In the Wi-Fi transmission condition, BLE communication and sensor acquisition were disabled while *TinZr* connected to a Wi-Fi network and transmitted User Datagram Protocol (UDP) packets at 100 Hz. UDP was selected because it enabled continuous packet transmission at a fixed rate, providing a consistent and repeatable Wi-Fi workload for current consumption measurements. This condition represents the current required for active Wi-Fi-based wireless data transmission without physiological signals sensing.

Table 2 summarizes the average current drawn and the estimated runtime for the above-listed operating modes. Using a 150 mAh Li-Po battery, *TinZr* is capable of acquiring and wirelessly transmitting continuous multi-modal physiological data from connected sensing modules, which is desirable for research applications requiring physiological or other sensor measurements. The battery capacity should be selected based on the expected recording duration, as longer sessions may require a higher-capacity battery.

TinZr also supports battery-voltage monitoring through the *bat-lvl* input, allowing users to implement a low-battery warning or automatic recording stop in the firmware. The battery used in this study also includes protection circuitry that disconnects the battery near the end of discharge [28].

The 3.3 V rail is generated by an AP2112K-3.3TRG1 low-dropout regulator, which has a maximum output current of 600 mA. Because this rail also powers the ESP32-C3 and onboard peripherals, the current available to external sensors depends on the operating configuration. More current may be available when power-intensive functions such as Wi-Fi or BLE are not used. As a conservative recommendation, we suggest limiting the combined continuous external load to 50 mA. Higher external loads may be supported depending on the active *TinZr* functions and the overall system configuration, while sensors requiring more current than *TinZr* can provide may require a separate external regulated power source.

7.5. Wireless communication characterization

Wireless communication performance was characterized to evaluate the practical reliability of *TinZr* BLE data transmission as an integrated wearable system. Although BLE communication is provided by the ESP32-C3 microcontroller, the practical wireless performance of *TinZr* also depends on the board layout, antenna placement, enclosure, firmware implementation, receiver hardware,

Table 2

Measured battery current consumption and estimated battery life under representative *TinZr* operating modes. Estimated runtime was calculated for a 150 mAh single-cell Li-Po battery.

Mode	Sensor configuration	Sampling rate	Battery current	Estimated runtime
Sleep Idle	None	–	0.376 mA	16.6 days
Active Idle	None	–	14.8 mA	10.14 h
BLE Advertising	None	–	42.2 mA	3.55 h
BLE Streaming	None	50 Hz	40.8 mA	3.68 h
BLE Streaming	None	100 Hz	41.1 mA	3.65 h
BLE Streaming	None	200 Hz	41.6 mA	3.61 h
BLE + MAX30102 ^a	Red and infrared PPG	100 Hz	47.9 mA	3.13 h
BLE + MAX30102	Red and infrared PPG	200 Hz	51.7 mA	2.90 h
BLE + MAX30001 ^b	Single ECG lead	128 Hz	50.2 mA	2.99 h
BLE + MAX30001	Single ECG lead	256 Hz	51.7 mA	2.90 h
BLE + MAX30102 + MAX30001	100 Hz/128 Hz	100 Hz	56.5 mA	2.65 h
BLE + MAX30102 + MAX30001	200 Hz/256 Hz	200 Hz	61.0 mA	2.46 h
Wi-Fi Transmission	None	UDP, 100 Hz	93.0 mA	1.61 h

^a We collect red and infrared PPG signals are collected from the MAX30102 sensor.

^b We collect a single lead ECG waveform from a custom MAX30001-based board.

Table 3

Practical BLE wireless connectivity characterization of *TinZr*. All tests were performed for 180 s. Single-device tests were performed at 100 Hz, and multi-device tests were performed using five *TinZr* devices at 10 Hz per device.

Condition	No. Devices	Distance (ft)	Sampling rate (Hz)	Packets expected	Packet loss (%)	RSSI (dBm)
Direct LOS	1	< 1 to 20	100	18,000	0	–59.93 to –89.54
One Door	1	< 1 to 20	100	18,000	0	–62.89 to –88.80
Direct LOS	5	< 1 to 20	10	9000	0	–68.03 to –87.75
One Door	5	< 1 to 20	10	9000	0	–68.41 to –88.04

and operating environment. Therefore, the goal of this experiment was not to revalidate the ESP32-C3 wireless specification, but rather to quantify the functional BLE performance of *TinZr* under representative real-life conditions.

Two BLE characterization tests were performed. The first test evaluated single-device high-rate transmission, typically used for physiological data acquisition from a wearable sensor. The second test evaluated simultaneous multi-device coordination, used in applications where multiple *TinZr* devices are used together, such as collaborative robotics.

For both tests, *TinZr* devices continuously transmitted BLE data packets to a laptop receiver running a custom Python application. Each transmitted BLE packet contained a device identifier, packet counter, timestamp, and representative dummy sample frames to emulate continuous data streaming. The receiver logged all received packets and used the packet counter sequence to identify missing data. Packet loss was calculated as the difference between the expected number of packets and the number of packets received, divided by the expected number of packets and expressed as a percentage. The received signal strength indicator (RSSI) was also recorded to provide a practical measure of the signal's quality. All wireless tests were performed for 180 s, and the reported results represent the average values over the test duration. A detailed description of the two tests is as follows:

- (1) In the single-device test, one *TinZr* device transmitted continuously at 100 Hz; a rate suitable for physiological data acquisition from wearable sensors. The device was placed at distances of <1 ft, 5 ft, 10 ft, 15 ft, and 20 ft from the laptop receiver. Measurements were performed under direct line-of-sight (LOS) conditions and with one-door separation, where the *TinZr* was positioned behind a wooden door relative to the receiver. This test evaluated whether *TinZr* could maintain reliable high-rate BLE transmission over practical indoor distances.
- (2) In the multi-device test, five *TinZr* devices transmitted simultaneously to the same laptop receiver at 10 Hz per device using the same BLE packet format used in the first test. The lower per-device rate was selected to represent multi-*TinZr* coordination scenarios, such as collaborative or swarm robotics, where each *TinZr* transmits lower-rate state information such as position, motion, or status rather than continuous high-rate physiological waveforms. Multi-device tests were performed under similar distances and conditions.

Across the single-device test, *TinZr* maintained continuous BLE transmission with no observed packet loss for all distances (from <1 ft to 20 ft) under both direct LOS and one-door conditions. RSSI decreased with increasing distance and obstruction, as expected, but remained sufficient to maintain the BLE connection and data stream throughout the 180 s trials. In the multi-device test, the five *TinZr* devices were able to stream simultaneously to the laptop receiver, demonstrating practical multi-node operation under the tested conditions. Table 3 summarizes the practical BLE characterization results, including the number of *TinZr* devices, distance, sampling rate, expected packets, packet loss, and RSSI.

CRediT authorship contribution statement

Ludvik Alkhoury: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Tony Moore:** Conceptualization, Resources, Visualization, Writing – review & editing. **Petrás Swissler:** Resources, Visualization, Writing – review & editing. **N. Jeremy Hill:** Resources, Visualization, Writing – review & editing. **Sudhin A. Shah:** Resources, Visualization, Writing – review & editing. **Moshe Kam:** Resources, Visualization, Funding acquisition, Supervision, Writing – review & editing.

Ethics statement

The data collection from the participant reported in Section 7.2 was approved by Weill Cornell Medicine Institutional Review Board (WCM-IRB) (IRB number 19-05020070). The study was conducted in accordance with the local legislation and institutional requirements. Written informed consent was obtained by the participants' legal guardians/next of kin.

Declaration of competing interest

Ludvik Alkhoury, N. Jeremy Hill, and Sudhin A. Shah hold equity in and serve as the founding team for Cognitive Signals Inc. These interests have been disclosed to the authors' respective institutions and are managed in accordance with their conflict-of-interest policies. The remaining authors declare no competing interests.

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